Plant Vogtle Nuclear Power Plant
A Comprehensive Case Study

America's Largest Nuclear Power Station —
Cost Overruns, Schedule Delays, and Lessons Learned

Based on the Following Key Studies:

1. **"The Truth About Vogtle"**
   Patty Durand, Kim Scott, Glenn Carroll
   Sponsored by: Cool Planet Solutions, Georgia WAND, Nuclear Watch South, GCV Education Fund, Center for a Sustainable Coast, Concerned Ratepayers of Georgia (2023–2024)
2. **"Ratepayer Robbery — The True Cost of Plant Vogtle"**
   Georgia Conservation Voters & Environment Georgia
   December 2021
3. **"Southern Company's Troubled Vogtle Nuclear Project"**
   David Schlissel, Director of Resource Planning Analysis
   Institute for Energy Economics and Financial Analysis (IEEFA), January 2022
4. **"Vogtle Units 3 & 4: Twenty-Third Semi-Annual Construction Monitoring Report"**
   Georgia Power Company / Southern Nuclear Operating Company
   Docket No. 29849, August 2020
5. **"Final Report SJR 79: Kentucky Nuclear Energy Development"**
   Kentucky Office of Energy Policy
   Report to Kentucky Legislative Research Commission, November 17, 2023
6. **"Commercial Nuclear Power Reactors — Operating Reactors"**
   U.S. Nuclear Regulatory Commission (NRC)
   Document No. ML25051A096 (Appendix A, NRC Reactor List)

Research Analysis Report
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Introduction

Georgia Power's Plant Vogtle, located in Burke County, Georgia near Augusta, is the largest nuclear power station in the United States. It is the only site in the country where new commercial nuclear reactors have been constructed in over 30 years, making it a landmark case study in both nuclear energy policy and large infrastructure project management.

Plant Vogtle Units 1 and 2 came online in 1987 and 1989, respectively, having themselves experienced dramatic cost overruns—escalating from an initial estimate of $660 million to a final cost of $8.8 billion. Despite this precedent, Georgia Power Company, a wholly-owned subsidiary of the Southern Company, sought and received approval to construct two additional reactors (Units 3 and 4) employing Westinghouse's new AP1000 passive-safety reactor design.

This case study synthesizes findings from six key reports and documents produced between 2020 and 2025 by a range of organizations—from advocacy groups and independent financial analysts to regulatory agencies and utility operators—to provide a comprehensive account of Plant Vogtle's development, cost, controversy, and implications for U.S. energy policy.

Plant Vogtle: Overview and Background

Location and Ownership

Plant Vogtle is situated on the Savannah River in Burke County, Georgia. The four owners of Vogtle Units 3 and 4 are:

- **Georgia Power Company** (lead owner and agent) — a subsidiary of Southern Company
- **Oglethorpe Power Corporation**
- **Municipal Electric Authority of Georgia (MEAG)**
- **City of Dalton, Georgia**

Southern Nuclear Operating Company (SNC), acting as agent for Georgia Power, managed the construction and operations of Units 3 and 4.

The AP1000 Reactor Design

Units 3 and 4 use Westinghouse's AP1000 reactor design, a pressurized water reactor (PWR) with passive safety systems. When Georgia Power applied to the Georgia Public Service Commission (PSC) in 2008 for certification, it argued that the AP1000's modular construction techniques would enable the project to be completed faster and at lower cost than prior nuclear builds. This claim proved to be significantly overstated.

Historical Context: Units 1 and 2

The construction history of Vogtle Units 1 and 2 provided a clear warning sign that was ultimately ignored. Initial cost estimates for Units 1 and 2 stood at $660 million; the final cost ballooned to $8.8 billion by the time they came online in 1987 and 1989—an overrun exceeding 1,200%. A U.S. Department of Energy study of 75 reactors built between 1966 and 1977 found average construction costs ran 207% over initial estimates. The historical record provided ample evidence that new nuclear construction was prone to severe cost escalation.

Construction Timeline and Schedule Delays

Original Schedule

Georgia Power proposed the following commercial operation dates (CODs) when seeking certification in 2008:

Unit	Originally Estimated COD	Actual / Revised COD
Unit 3	April 1, 2016	July 2023
Unit 4	April 1, 2017	Late 2023 / Early 2024

Both units entered commercial operation more than **six years behind schedule**.

Key Construction Milestones and Setbacks

The Twenty-Third Semi-Annual Vogtle Construction Monitoring Report (August 2020, Docket No. 29849) covered the period January 1 to June 30, 2020 and reported:

- Southern Nuclear Operating Company received no NRC Notices of Violation during the reporting period and maintained a **green (favorable) status** under the NRC's Construction Reactor Oversight Process (cROP).
- The site sustained more than **34 million man-hours** without a lost-time incident before an incident occurred in April 2020.
- The **COVID-19 pandemic** caused significant workforce disruptions in 2020, including spikes in infections (over 800 workers tested positive), workforce reductions, and increased costs and productivity challenges.
- Despite pandemic-related setbacks, essential construction activities including the Passive Containment Cooling Water Storage Tank (CB20) on the Unit 3 Shield Building continued.

Westinghouse Bankruptcy (2017)

The most dramatic event in Vogtle's construction history was the March 2017 bankruptcy filing by Westinghouse Electric Company, the contractor responsible for engineering, procurement, and construction. The bankruptcy was triggered in part by billions of dollars in losses on the Vogtle and V.C. Summer nuclear construction projects.

In South Carolina, an identical AP1000 project (SCANA's V.C. Summer) was abandoned in July 2017 following nine consecutive rate increases. Investigations revealed that Westinghouse and utility executives had misrepresented project progress and costs, ultimately resulting in criminal charges and jail time for two SCANA executives.

In Georgia, the PSC voted to continue the Vogtle project despite the Westinghouse bankruptcy, without imposing meaningful consumer protections against further overruns.

Cost Analysis and Financial Impact

Cost Escalation

Year	Estimated Total Cost	Source
2009 (original approval)	$14.0 billion	Georgia PSC Certification
2017 (post-bankruptcy)	$25+ billion	Georgia PSC Proceeding
2021	$32+ billion	GCV Ratepayer Robbery Report
2022	$30+ billion	IEEFA / Schlissel Report

The total cost of Units 3 and 4 more than **doubled** from the original estimate of $14 billion to above $30–32 billion, making Plant Vogtle the most expensive power plant ever built on Earth at the time of its completion.

Cost Compared to Alternatives

The GCV "Ratepayer Robbery" report (December 2021) calculated that the cost per megawatt-hour of energy from Vogtle was **6 to 9 times higher** than comparable energy from renewable sources or natural gas that could have produced the same 2,200 MW of generating capacity.

The Nuclear Construction Cost Recovery (NCCR) Tariff

Georgia was one of a small number of states with a "Construction Work in Progress" (CWIP) law allowing utilities to collect construction costs from ratepayers *before* a plant enters commercial operation. The NCCR tariff allowed Georgia Power to collect revenues from customers throughout the construction period, effectively shifting financial risk from shareholders to ratepayers. Between 2009 and 2021, Georgia Power collected multiple NCCR-based rate increases from its customers.

Return on Equity and Regulatory Capture

Critics, including the GCV report, argued that the Georgia PSC allowed Georgia Power one of the **highest returns on equity (ROE)** of any utility in the United States. This structure meant Georgia Power *profited from construction delays*: longer construction timelines increased the capital base on which the utility earned its guaranteed rate of return. This created a perverse incentive that contributed to inadequate oversight of project management.

Regulatory Oversight and the Georgia PSC

Role of the Georgia Public Service Commission

The Georgia Public Service Commission (PSC) is the body responsible for regulating Georgia Power as a monopoly utility and for protecting ratepayers. The PSC approved the

Vogtle project in March 2009 and voted to continue construction after the Westinghouse bankruptcy in 2017.

Critics across multiple reports characterized the PSC as suffering from "**regulatory capture**"—a phenomenon where a regulatory body acts in the interests of the industry it regulates rather than the public. Georgia Power itself referred to this relationship as "constructive regulation."

Key PSC Decisions

- **March 2009**: Approved Georgia Power's application to build Units 3 and 4.
- **2017**: Voted to continue construction despite Westinghouse bankruptcy and available cancellation option; declined to impose consumer protections.
- **December 2023**: Final PSC vote following completion of the project, amid ongoing protests from ratepayer advocacy groups.

NRC Oversight

The U.S. Nuclear Regulatory Commission (NRC) monitored construction through its Construction Reactor Oversight Process (cROP). Per the Twenty-Third VCM Report (August 2020), Southern Nuclear Operating Company maintained a green (satisfactory) NRC performance indicator throughout 2020.

The NRC document ML25051A096 lists Plant Vogtle among the United States' licensed commercial nuclear power reactors, confirming its operational status in the NRC's national reactor registry.

Study-by-Study Analysis

Study 1: "The Truth About Vogtle"

Authors: Patty Durand (Cool Planet Solutions), Kim Scott (Georgia WAND), Glenn Carroll (Nuclear Watch South)
Sponsors: Center for a Sustainable Coast; Concerned Ratepayers of Georgia; Cool Planet Solutions; GCV Education Fund; Georgia WAND; Nuclear Watch South

This report, the most comprehensive advocacy document in the repository, presents a critical examination of the entire Vogtle project from initial approval through completion. Key findings include:

- Vogtle 3 and 4 are the **only new nuclear reactors built in the U.S. in over 30 years**.
- Cost overruns began immediately after construction started in 2009.
- The Westinghouse bankruptcy in March 2017 created a clear decision point that the Georgia PSC failed to use to protect ratepayers.
- The report systematically debunks seven common myths about nuclear energy:
    1. Nuclear energy is clean

2. Nuclear energy is safe
3. Nuclear waste is "no big deal"
4. Small Modular Reactors (SMRs) are different
5. Nuclear is required to back up renewables
6. Nuclear is required to meet future growth
7. Nuclear is needed to combat climate change

- The report documents **scandals and litigation** surrounding the project, including parallels with the V.C. Summer SCANA scandal in South Carolina where executives faced criminal charges.
- The U.S. South is described as the "nuclear hub of the United States," hosting over three dozen operational reactors, weapons manufacturing complexes, and nuclear fuel factories.

Study 2: "Ratepayer Robbery — The True Cost of Plant Vogtle"

Authors: Georgia Conservation Voters & Environment Georgia
Date: December 2021

This report frames the Vogtle project as a financial harm imposed on Georgia ratepayers. Key findings include:

- Plant Vogtle is described as "**the most expensive power plant ever built on Earth**," with costs exceeding $32 billion for 2,200 MW of capacity.
- The cost-per-megawatt-hour from Vogtle is 6–9 times higher than renewable alternatives.
- Georgia Power's system outages **consistently exceeded U.S. averages** for investor-owned utilities, undermining the utility's claim of "superior service."
- The report details the NCCR tariff mechanism and documents how Georgia Power collected rate increases from customers during construction.
- The "Parental Guarantee" from Southern Company is analyzed, showing the limits of shareholder protection for ratepayers.
- The report argues that the PSC's 2017 decision to continue construction was the **costliest regulatory failure** in Georgia history.
- It concludes that cancellation and replacement with renewables in 2017 would have been cheaper for customers.

Study 3: "Southern Company's Troubled Vogtle Nuclear Project"

Author: David Schlissel, Director of Resource Planning Analysis, IEEFA
Date: January 2022

This report, produced by the Institute for Energy Economics and Financial Analysis (IEEFA), provides an independent financial and engineering assessment. Key findings:

- Total cost had climbed above $30 billion by January 2022, with completion still expected to be **more than six years behind the original schedule**.
- Georgia Power's 2008 application claimed modular AP1000 construction would reduce both cost and schedule—a claim that proved false.
- The report draws a direct line between the failure of Vogtle 1 & 2 (which cost $8.8 billion versus an estimate of $660 million) and the failure of Units 3 & 4, arguing these outcomes were **predictable** given nuclear construction history.
- A DOE study of 75 reactors found average cost overruns of 207%—evidence that was available to the PSC before it approved Vogtle 3 & 4.
- Commercial operations for Unit 3 were projected at 2022 and Unit 4 at 2023, though actual completion slipped further.

Study 4: "Vogtle Units 3 & 4: Twenty-Third Semi-Annual Construction Monitoring Report"

Issuer: Georgia Power Company / Southern Nuclear Operating Company
Date: August 2020 (covering January–June 2020)
Docket: No. 29849

This is an official regulatory filing by the utility itself, submitted to the Georgia PSC. Key contents:

- The report describes the overall safety and quality management culture on site, including a commitment to "**safety first**."
- As of June 2020, SNC maintained green status under the NRC's cROP and had received no Notices of Violation.
- COVID-19 significantly impacted the project: over 800 workers tested positive, and workforce reductions in April 2020 caused productivity losses, cost increases, and milestone delays.
- The report shows the Passive Containment Cooling Water Storage Tank (CB20) atop the Unit 3 Shield Building being completed—a visible symbol of construction progress during a turbulent period.
- This report represents the utility's perspective, emphasizing compliance and safety culture, in contrast to the critical perspective of the advocacy reports.

Study 5: "Final Report SJR 79 — Kentucky Nuclear Energy Development"

Issuer: Kentucky Office of Energy Policy
Date: November 17, 2023
Submitted to: Kentucky Legislative Research Commission

While primarily focused on Kentucky's nuclear energy future, this report provides important national context drawn in part from the Vogtle experience:

- Kentucky currently has **no operational commercial nuclear reactors**, having relied on coal and natural gas historically.

- The report analyzes the **value proposition of nuclear energy** in terms of reliability, baseload power, and decarbonization.
- It examines barriers to nuclear development in Kentucky, including:
  - High capital costs and financing risk (informed by Vogtle's experience)
  - Workforce and supply chain gaps
  - Regulatory and permitting uncertainty
  - Public perception challenges
- The report recommends creating a **Kentucky Nuclear Energy Development Authority** with near-term goals (0–3 years) and intermediate-term goals (3+ years).
- It identifies Small Modular Reactors (SMRs) as a "gated opportunity" for future deployment, contingent on the success of national SMR demonstration projects.
- The Vogtle project's lessons—particularly the dangers of cost overruns with large light-water reactors—are implicitly used to justify a cautious, phased approach to nuclear development in Kentucky.

Study 6: NRC Document ML25051A096 — "Commercial Nuclear Power Reactors: Operating Reactors"

Issuer: U.S. Nuclear Regulatory Commission (NRC)
Document: Appendix A, Operating Reactors List

This NRC document places Plant Vogtle in the national context of licensed and operating commercial nuclear power reactors in the United States. Key information:

- The document lists all NRC-licensed operating reactors by plant name, licensee, location, docket number, reactor type (e.g., PWR-DRYAMB, BWR), NSSS supplier, architect-engineer, constructor, net MWe capacity, and NRC Region.
- Plant Vogtle appears as a licensed facility in NRC Region II (Southeast).
- The listing confirms that Plant Vogtle Units 3 and 4 are operational, using the AP1000 (PWR) design supplied by Westinghouse.
- The document contextualizes Vogtle among a national fleet that includes: Arkansas Nuclear One (Entergy), Beaver Valley Power Station (Energy Harbor/Vistra), Braidwood Station (Constellation), and dozens of others.
- The NRC's role as licensing authority—distinct from the state PSC's economic regulation—is clarified by this document.

Cross-Cutting Themes and Lessons Learned

Theme 1: The Persistent Pattern of Nuclear Cost Overruns

Every independent study reviewed here confirms that large-scale nuclear construction in the United States has been associated with dramatic cost overruns. From Vogtle

Units 1 & 2 in the 1980s to Units 3 & 4 in the 2010s–2020s, and across other projects documented by the DOE and NRC, cost escalation appears to be a structural feature of nuclear construction rather than an anomaly.

Theme 2: Regulatory Capture and Ratepayer Risk

Multiple reports identify a pattern of regulatory capture at the Georgia PSC. The NCCR tariff mechanism enabled Georgia Power to pass construction costs to ratepayers in advance, removing the market discipline that would otherwise incentivize cost control. The PSC's failure to impose consumer protections following the Westinghouse bankruptcy in 2017 is highlighted by all three independent studies as a critical governance failure.

Theme 3: The V.C. Summer Comparison

The abandoned V.C. Summer project in South Carolina serves as a counterfactual throughout the advocacy reports. Had Georgia followed South Carolina's decision to cancel the project in 2017, ratepayers would have avoided billions in additional costs. The criminal accountability for SCANA executives in South Carolina—absent in Georgia—is cited as evidence of differential regulatory enforcement.

Theme 4: Implications for Future Nuclear Development

The Kentucky SJR 79 report illustrates how Vogtle's troubled history is shaping nuclear policy discussions in other states. The report's cautious framing of SMRs as a "gated opportunity" reflects lessons drawn from Vogtle: the importance of proven technology, phased investment, and independent financial oversight before committing to large capital expenditures.

Theme 5: COVID-19 and Infrastructure Resilience

The VCM Report from August 2020 documents how the COVID-19 pandemic exposed the vulnerability of large construction projects to workforce disruptions. The virus caused schedule delays and cost increases at Vogtle, adding to an already troubled project. This experience points to the need for resilient workforce planning and contingency budgets in large infrastructure projects.

Key Data Summary

Parameter	**Data**
Location	Burke County, Georgia, USA (near Augusta, on the Savannah River)
Plant Type	Nuclear Power Plant (Pressurized Water Reactor — AP1000)
Units 3 & 4 Capacity	2,200 MW (combined)
Owner (Lead)	Georgia Power Company (subsidiary of Southern Company)
Other Owners	Oglethorpe Power, MEAG, City of Dalton

Parameter	Data
Reactor Supplier	Westinghouse Electric Company (AP1000 design)
NRC Region	Region II (Southeast)
Original Cost Estimate	\$14 billion (2009 PSC application)
Final Cost Estimate	\$32+ billion
Cost Overrun	~130% above original estimate
Original COD (Unit 3)	April 1, 2016
Actual COD (Unit 3)	July 2023
Schedule Delay	6+ years behind schedule
Westinghouse Bankruptcy	March 2017
NRC cROP Status (2020)	Green (Satisfactory)
Man-Hours without LTI	34+ million (as of April 2020)
COVID-19 Impact	800+ positive tests on site; workforce reduction; cost increases
Cost vs. Renewables	6–9× higher per MWh than renewable alternatives

Conclusion

Plant Vogtle Units 3 and 4 represent the most significant nuclear energy construction project in the United States in a generation. As the only new commercial nuclear reactors built in the country in over 30 years, they have served as both an ambition and a cautionary tale.

The six studies reviewed in this case study converge on several conclusions:

1. **Cost overruns were predictable.** Historical data from Vogtle 1 & 2 and the broader U.S. nuclear fleet clearly indicated that the original \$14 billion estimate was unrealistic. The project ultimately cost more than double that amount.

2. **Regulatory oversight failed ratepayers.** The Georgia PSC, functioning under a model critics describe as regulatory capture, repeatedly made decisions that protected utility profits over consumer interests—most critically in 2017 when the Westinghouse bankruptcy created a legitimate off-ramp.

3. **The financial burden fell on ratepayers.** Through the NCCR tariff and repeated rate increases, Georgia electricity customers funded the project during construction and will continue to pay elevated rates for decades.

4. **Alternatives existed and were not chosen.** Multiple analyses demonstrate that renewable energy and natural gas alternatives could have provided equivalent

capacity at a fraction of the cost, especially given the dramatic price reductions in solar and wind between 2009 and 2023.

5. **Vogtle is now shaping national nuclear policy.** The project's completion, despite its cost and schedule failures, has been cited by nuclear advocates as proof of concept for AP1000 technology. Other states such as Kentucky are examining nuclear options while attempting to learn from Vogtle's mistakes through phased approaches and SMR technology.

Plant Vogtle stands as a complex legacy: technically completed but economically troubled; a proof of advanced reactor technology but also evidence of deep systemic failures in utility regulation, cost estimation, and infrastructure governance. Its lessons will shape U.S. energy policy for decades to come.

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[1] Durand, P., Scott, K., & Carroll, G. (n.d.). *The Truth About Vogtle.* Sponsored by Cool Planet Solutions, Georgia WAND, Nuclear Watch South, GCV Education Fund, Center for a Sustainable Coast, Concerned Ratepayers of Georgia.

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[4] Georgia Power Company / Southern Nuclear Operating Company. (August 2020). *Vogtle Units 3 & 4: Twenty-Third Semi-Annual Construction Monitoring Report.* Docket No. 29849. Submitted to Georgia Public Service Commission.

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[8] Georgia Public Service Commission. (August 1, 2008). *Georgia Power Company Application for the Certification of Vogtle Generating Units 3 and 4 and Upgraded Integrated Resource Plan.*